

Roadmap

This structured, step-by-step roadmap will take you from installation to writing your own programs.

Phase 1: Set Up Your Tools

Before writing code, you need an environment to run it

- **Install the JDK:** Download the latest [Oracle Java Development Kit \(JDK\)](#). JDK translates your code into a language your computer understands.
 - **Get an IDE:** Download an Integrated Development Environment (IDE) to write your code. [JetBrains IntelliJ IDEA \(Community Edition\)](#) is highly recommended for beginners. Alternative options include [Microsoft VS Code](#) or Eclipse.
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Phase 2: Learn Core Syntax

Start by learning how a basic program is structured. Every runnable Java program requires a primary entryway called the main method. Your initial checklist should include:

- **Variables:** Containers that hold data, like numbers or text.
 - **Data Types:** Differences between integers (int), decimals (double), characters (char), and true/false statements (boolean).
 - **Control Flow:** Using if/else conditions to make decisions.
 - **Loops:** Using for, while loops to repeat tasks automatically.
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Phase 3: Understand Object-Oriented Programming (OOP)

Java is fundamentally built on OOP. This means you organize your code into modular, reusable blocks.

Master these four pillars: [\[1, 2, 3\]](#)

- **Classes & Objects:** A class is a blueprint, and an object is the actual thing you build from it.
 - **Inheritance:** Letting a new class adopt the properties of an existing class.
 - **Polymorphism:** Allowing different classes to be treated as the same general type.
 - **Encapsulation:** Hiding data inside a class to keep it safe from outside interference.
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Phase 4: Advance Java Data structure and Data Handling & Lists

Programs need to manage groups of information. You will want to study:

- **Arrays:** Fixed-size lists used to store similar items.
- **ArrayLists:** Dynamic lists that automatically grow or shrink when you add or remove items.
- **HashMaps:** Key-value pairs used for rapid data lookups.